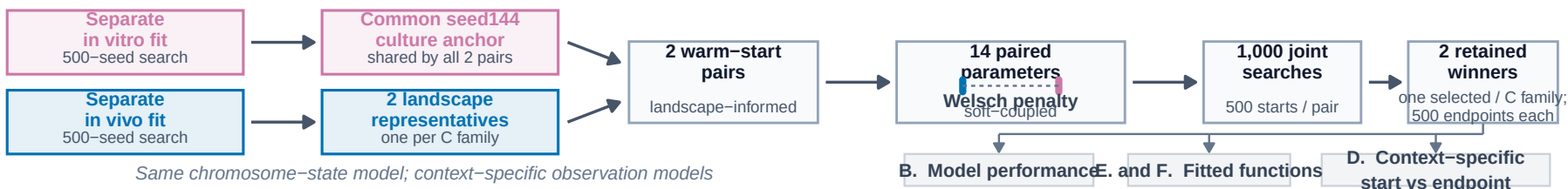


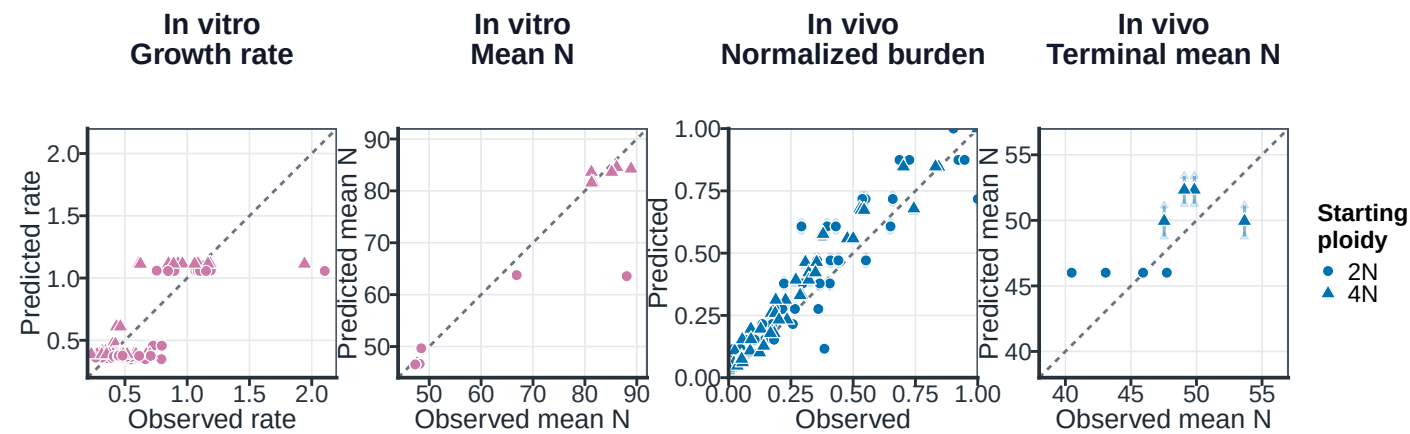
Joint-fit workflow, performance, mechanisms, and solution stability

2 primary-family pairs;
Panel D summarizes endpoint direction and within-family parameter-ratio stability.

A. Joint-fitting workflow

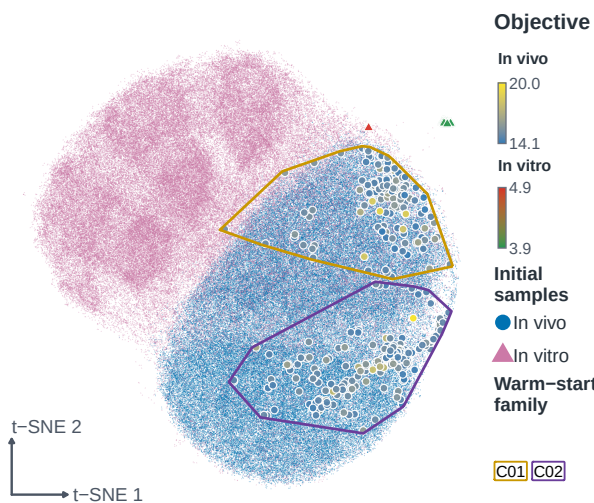


B. In-sample model performance

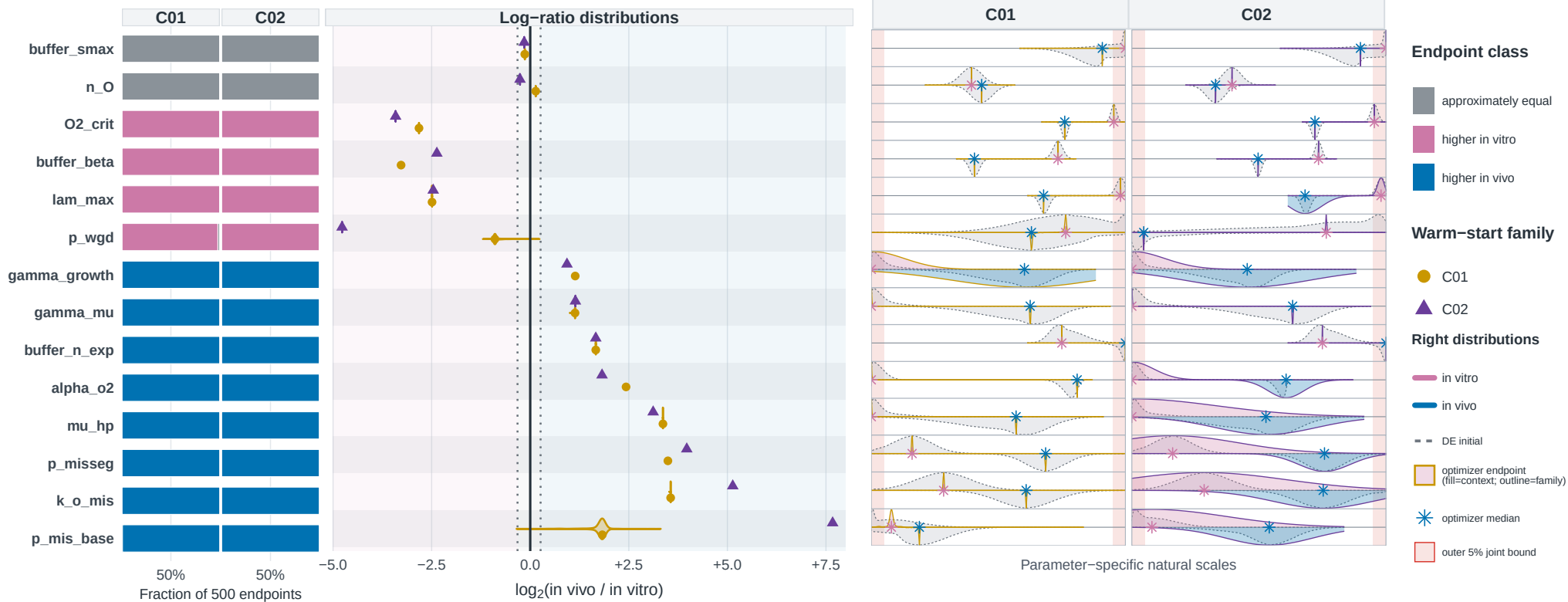


Circle: 2N start; triangle: 4N start. Pale: 2 selected fits; filled: median; bars: selected-fit range.

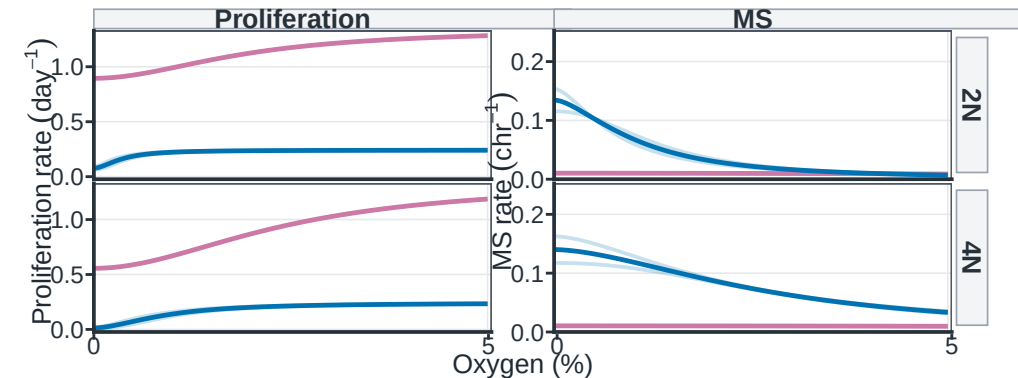
C. Warm-start map



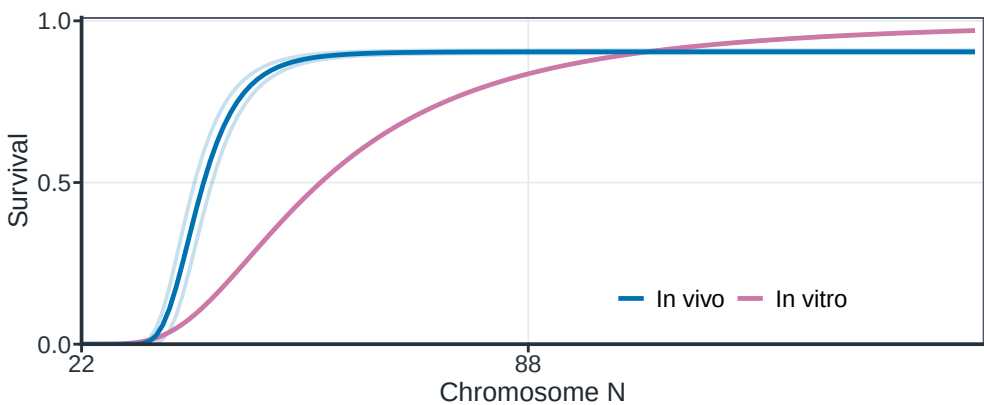
D. Joint-fit parameter stability



E. Oxygen-dependent fitted functions



F. Survival



B: circle = 2N start; triangle = 4N start. E-F: thin curves = 2 fits; thick curves = pointwise medians.
D: pink/gray/blue bars classify 500 optimizer endpoints per family and parameter;
center violins show non-degenerate endpoint log-ratio spread and symbols mark family medians;
right mirrored densities compare context-specific DE initial and optimizer-endpoint distributions.
The center D-axis spans the complete observed log-ratio range; right axes use parameter-specific natural scales.
C01, C02 retain the same primary-family identities across panels;
optimizer distributions are descriptive numerical-search summaries, not posterior or confidence distributions.